



Fig. 10: Commercial helium-neon laser power supplies



Fig. 11: Helium-neon laser with inbuilt power supply

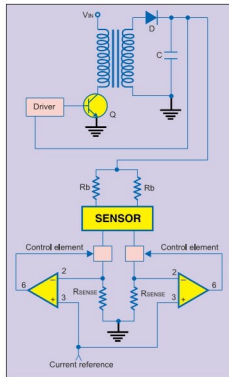


Fig. 12: Helium-neon power supply for ring laser gyro cavity

voltage to initiate the discharge. Fig. 8 shows a typical circuit. The power supply is configured around an exter-

nally driven fly-back type DC-to-DC converter. The ballast resistance is split into two resistances.

Initially, when there is no discharge, the voltage appearing after R_{SENSE} is equal to open circuit voltage. A fraction of this voltage is compared with V_{REF} and the comparator drives the high-voltage trigger circuit configured around SCR, R-C components and trigger transformer. As soon as discharge is struck, voltage appearing after R_{SENSE} drops and the drive pulses are withdrawn from SCR gate. The power supply provides the discharge current through series connection of resistors constituting ballast resistance.

In the third circuit topology, the converter circuit is designed to deliver an output power equal to the product of the required sustaining voltage at the power supply output and the discharge current. Since it is a constant-output-power converter, the output voltage would increase without limit in the ideal case in the event of zero current drawn from the power supply. When the output voltage exceeds the required initiating voltage, the plasma current drawn from the power supply forces the output voltage to fall to the sustaining voltage governed by the output power capability of the converter.

Compact helium-neon laser power supplies are commercially available both as AC-operated bench-top versions as well as DC-operated supplies for OEM market (Fig. 10). Helium-neon lasers with an integrated power supply are also available (Fig. 11).

Power supplies for ring laser gyroscope sensors

A ring laser gyroscope comprises a helium-neon ring laser cavity with two counter-propagating laser beams. A ring laser gyroscope, which is the heart of the inertial navigation systems of modern aircraft, missiles etc, makes use of a ring helium-neon laser in a closed cavity. Without going into details of its operational principle, it would suffice to mention here that stability of discharge current in each arm of the ring laser cavity is very crucial for proper functioning of the device. Also, it is important to have a stable value of difference between the currents in the two arms.

A power supply configuration that achieves this is shown in Fig. 12. As shown in the figure, there are two current-sense loops controlled by the same reference voltage. Another possible approach could be to use a precision reference for one arm and then use the actual controlled current flowing through that arm as the reference for controlling current in the other arm. The preferred circuit topology for the power supply is an externally-driven PWM-controlled fly-back converter generating a regulated output voltage equal to the sustaining voltage at the required plasma current.

Carbon dioxide laser power supplies

Carbon dioxide (CO_2) lasers are operated in both CW and pulsed-output modes. CW lasers are either DC-excited or RF-excited. Pulsed-output lasers are mainly transversely-excited atmospheric-pressure (TEA) lasers. Power supply circuits for DC-excited lasers are similar to the ones described in earlier paragraphs in the case of helium-neon lasers. Different power supply circuits discussed in the case of helium-neon lasers are equally valid for carbon dioxide lasers.

Present-day TEA CO_2 lasers use Marx bank capacitor systems. In this system, the arrangement of individual capacitors is such that they are in parallel during the charging process and in series during the discharge opera-